

eddy

Fostering interactive
learning in the classroom

ARTG 2400

Interaction Design Principles

Jayden, Olivia, Saanvi, Alissa





Agenda

1 Research

2 Ideation

3 Prototype & Demo

4 Bringing it into the Real World

5 Takeaways

1.1 SDG Research

Sustainable Development Goal:
Quality Education



4

Quality Education

Definition: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

(<https://sdgs.un.org/goals/goal4>)

01 Research

Our Findings

Initial Research

- Underfunded schools lack STEM programs and extracurriculars.
- Higher-income areas have more tech and enrichment access.
- Strict governments or outdated curricula limit quality education globally.

Personal Reflection

Jayden: Public school in Central New Jersey. Adequately funded with art, music, and computer skill learning programs

Olivia: Public elementary school in Brooklyn NY. Had limited STEM programs and was not super culturally diverse.

Alissa: Public school in Brookline, MA. Well-funded with some extra STEM and Arts programs.

Saanvi: Public school in Acton, MA. Well-funded music, STEM, debate programs. District-wide 1-to-1 program started in 2020.

01 Research

Our Findings

Initial Research

- Even in the U.S. there are disparities in quality of education
- With limits to technology in a technology driven country, children will be at a major disadvantage
- It is vital to spread these stories to children, so they can be aware of theirs and other's circumstances

Case Study - Peer Interview

Will Newkirk:

- Public School in Inner City Minneapolis, K-8
- Underfunded in the 2010's due to recession, other factors
- Did not have access to personal computers to practice typing and technological skills
- Prefers pencil and paper over computers
- Many of his peers are now in blue collar occupations, with little experience working with a computers



Our Findings

Interview: Saanvi's Mom –
Elementary Teacher,
[Boxborough, MA]

- We asked Saanvi's mom to share insights on how her students engage with technology both at school and at home.

Key Insights

- Tech integration varies widely based on individual student needs—some benefit greatly, while others are distracted or overstimulated by it.
- Ideal solution: parent-inclusive tech tools and cross-school collaborations for special needs education.
- Kids in the district tend to be pretty tech-proficient because of the area's higher income levels, but it really depends on the home environment and how involved their parents are.

1.2 Competitor Research

Discovery Education, Class Dojo, and Class.com

01 Research

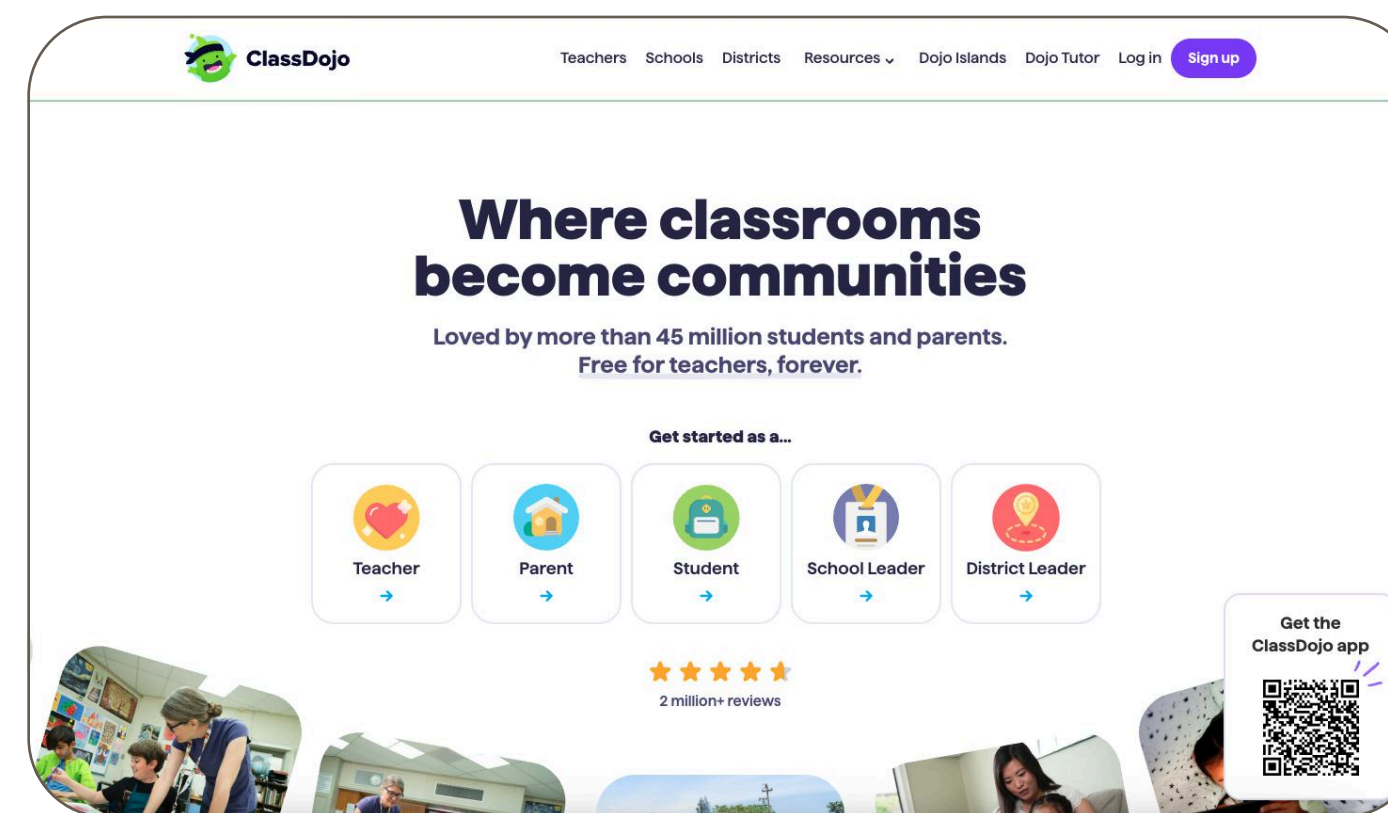
Our Findings

Discovery Education



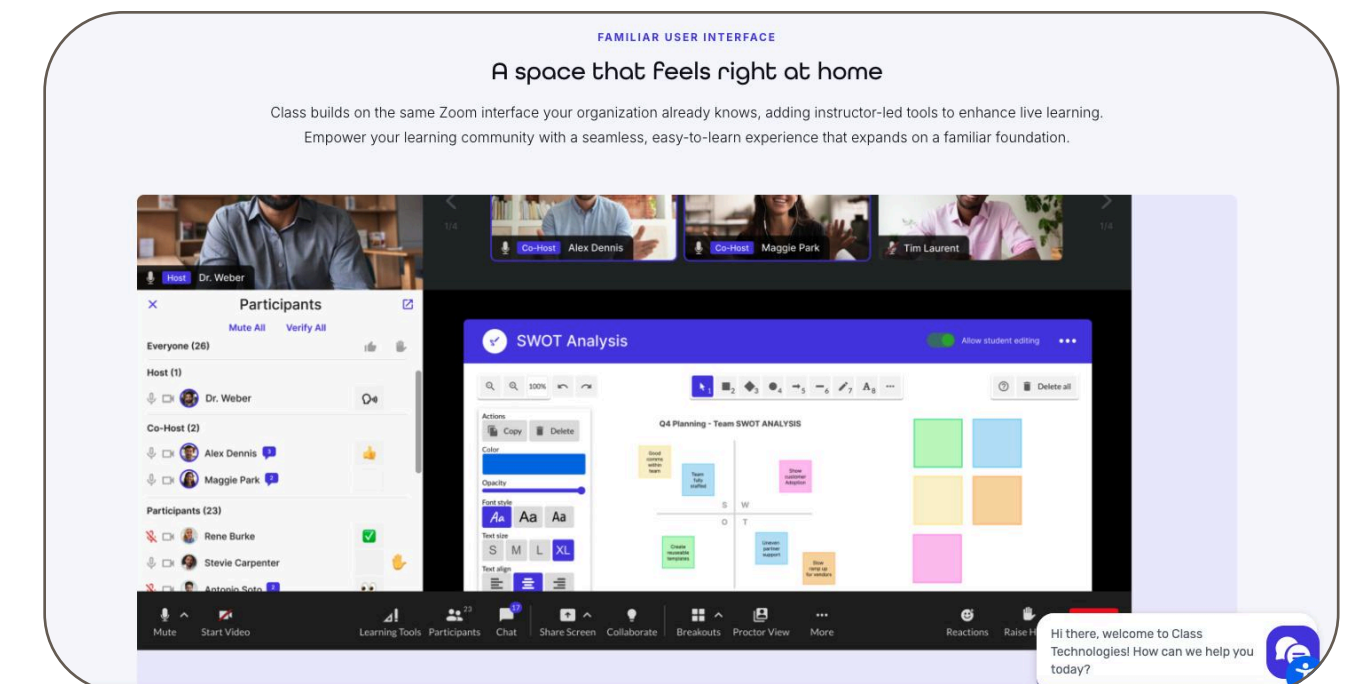
Educational service for personalized learning targeted for Elementary, Middle School - Administrator focused

Class Dojo



Keeping teachers, families, and kids connected

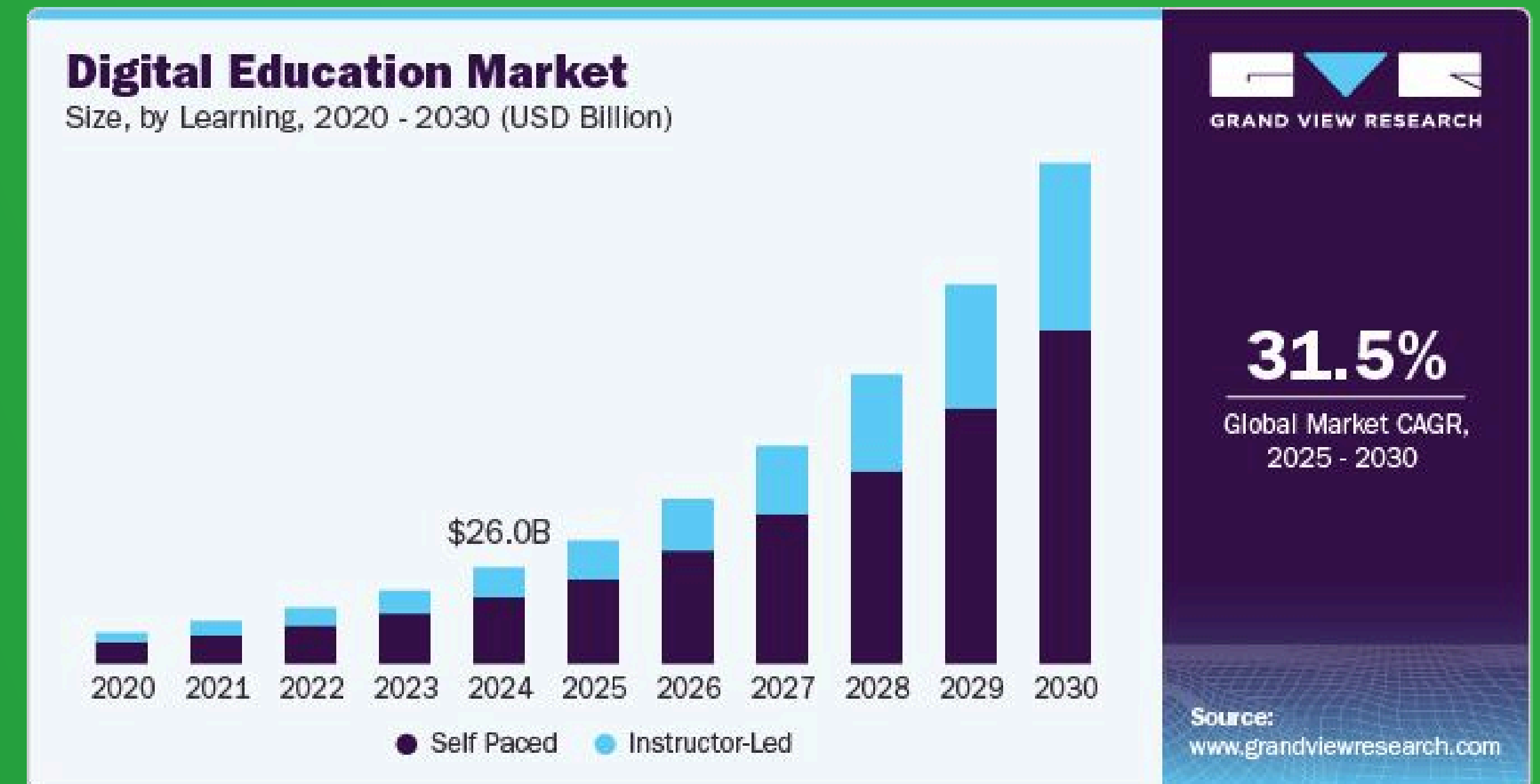
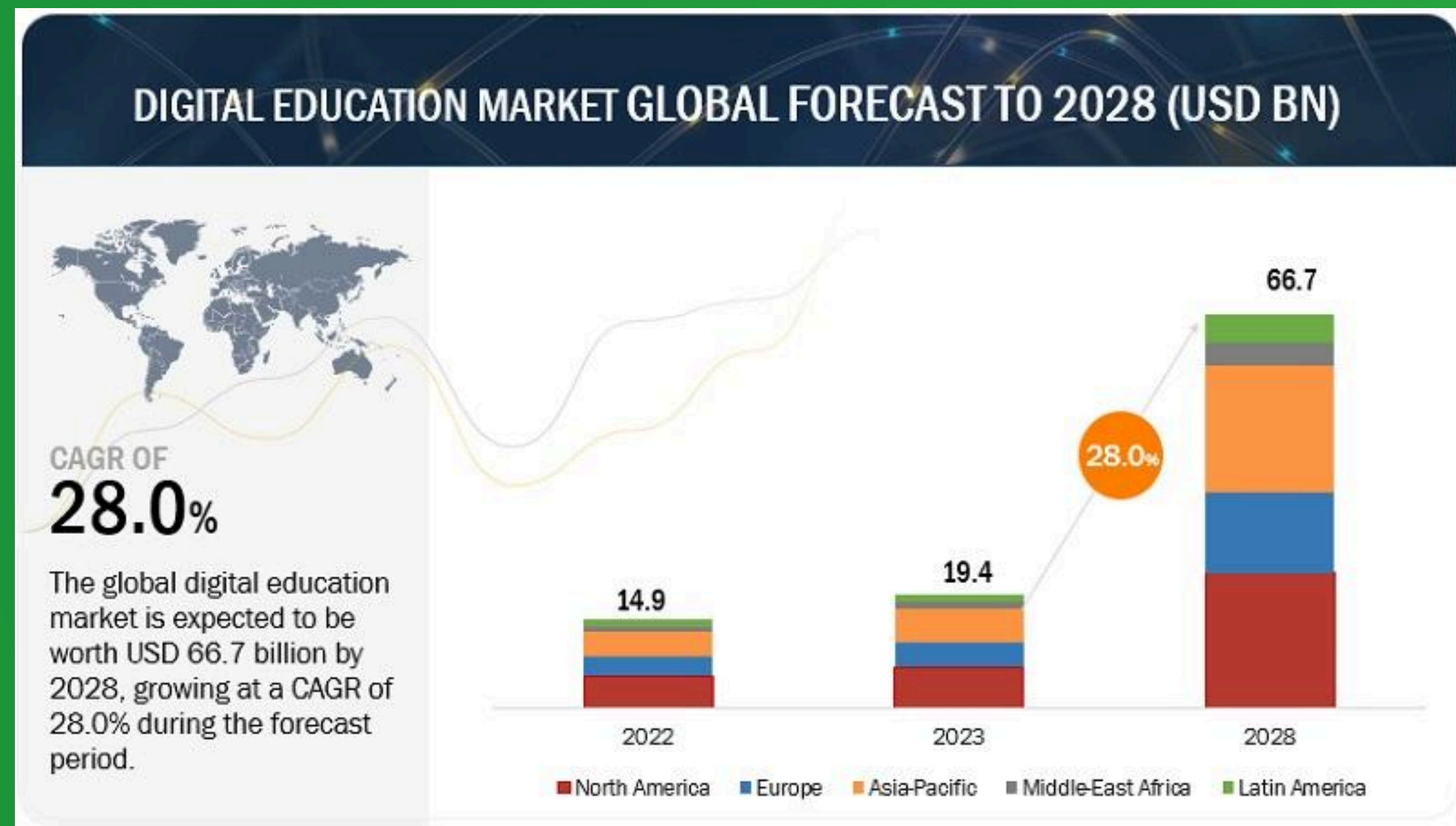
Class.com



The next generation of virtual classroom, built as an extension on Zoom

01 Research

Market Research



- The digital education market is expected to grow at a compound annual growth rate of around 28%, reaching 66.7 Billion by 2028.

Market Research

- The digital education market is forecasted to **grow exponentially** in the coming years, and it is imperative to implement as many resources for our children in order to fit the needs of the ever evolving education scene.
- \$109 annual subscription fee for competitor programs such as ClassDojo and Discovery Education.
- From Grand View Research:
 - **“North America dominated the global digital education market with revenue share of 32.4% in 2024.”**
 - **“The presence of robust IT & telecom industry... inclination in students towards adaptive learning are some of the key growth driving factors for this market.”**
- The self learning module of digital education is expected to grow drastically more than instructor led education. This shift demands attention, and marks a vital shift in education and planning for technologies that facilitate this.

01 Research

Personal Experiences

Jayden:

- Computer Labs with one computer for everyone,
- iPads in science class
- Personal Chromebooks starting Middle School

Olivia:

- Wide use of smart boards and projectors
- Library computers
- Did not have personal devices/computers until middle school

Alissa:

- Computer lab in elementary school
- Personal Chromebooks in Middle school but only used for in class activities (not take-home)
- High school introduced interactive whiteboards but not widely used

Saanvi

- Projectors and Interactive whiteboards
- Computer Labs. 'Chromebook cart'
- School library computers
- Personal Chromebooks starting during Covid

Problem Statement

How might we create **equitable learning opportunities** for **elementary school students** across different income levels in **Boston** by creating technology that encourages personalized learning?

02 Ideation

Coming up with ideas based on our research and problem statement



02 Ideation

some inspiration



initial feature ideas

video conference interface that provides real time interactive experiences for students working on projects together

Olivia Li

collection of educational projects or games users can choose from.

digital library

also provide options + resources that don't require internet access

Olivia Li

gamify features to make it engaging (prompts)

jayden

structured activities kids can do to get to know their virtual buddy better

students fill out a fun quiz that will allow them to get matched to another student based on interests and learning styles

Olivia Li

options for parent/teacher facilitation

Olivia Li

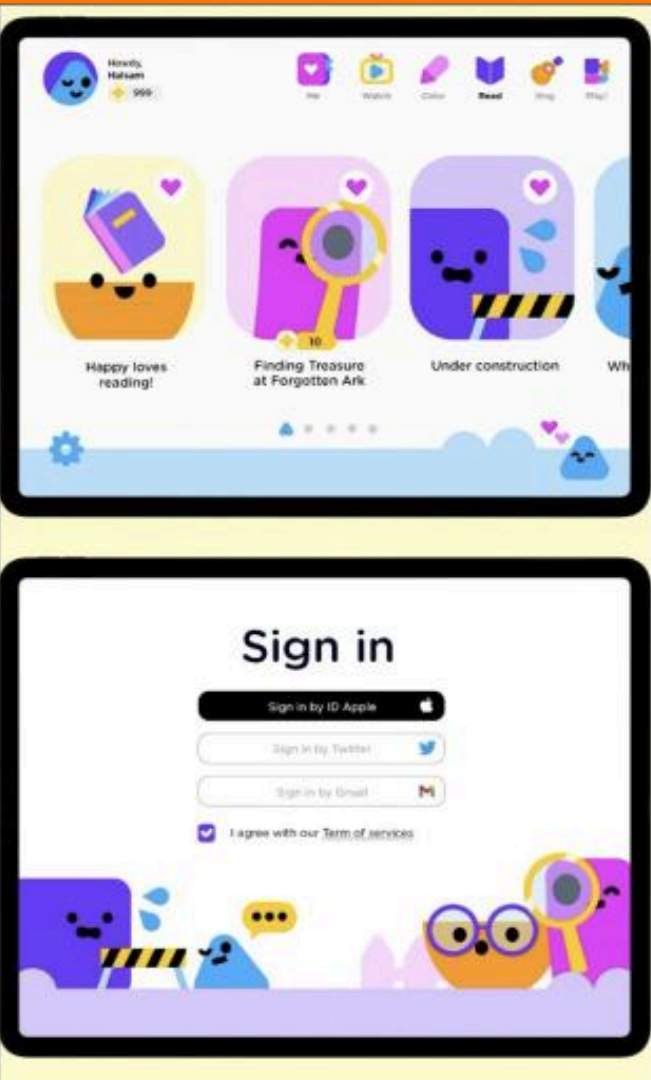
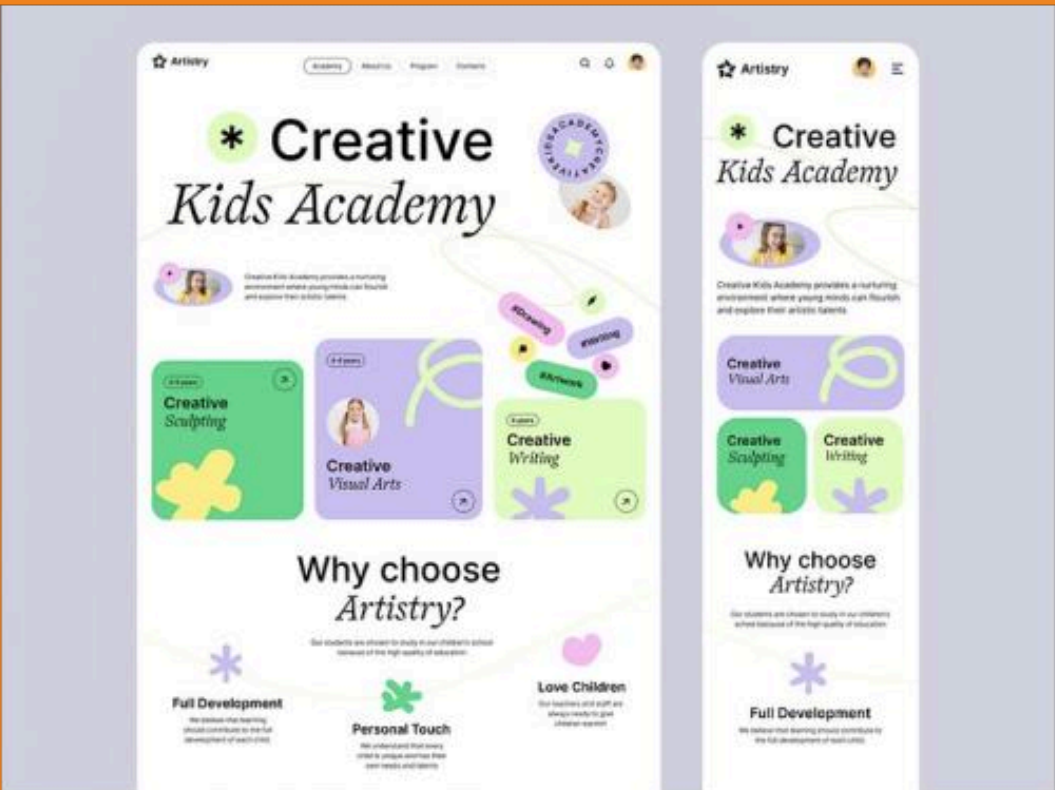
1. identify features
2. create user flows for each feature
3. lofi
4. hifi/prototype

Olivia Li

take inspiration from existing products while also making it more kid-friendly

- zoom, google meet, pbs kids, etc

Olivia Li



Defining Features

- Buddy/Pairing system through an Interest Survey
- Variety of educational activities
- Customization in profiles
- Gamification elements (gaining points, levels, etc.)
- Class roadmap

Logo, Branding, & Components

Typography

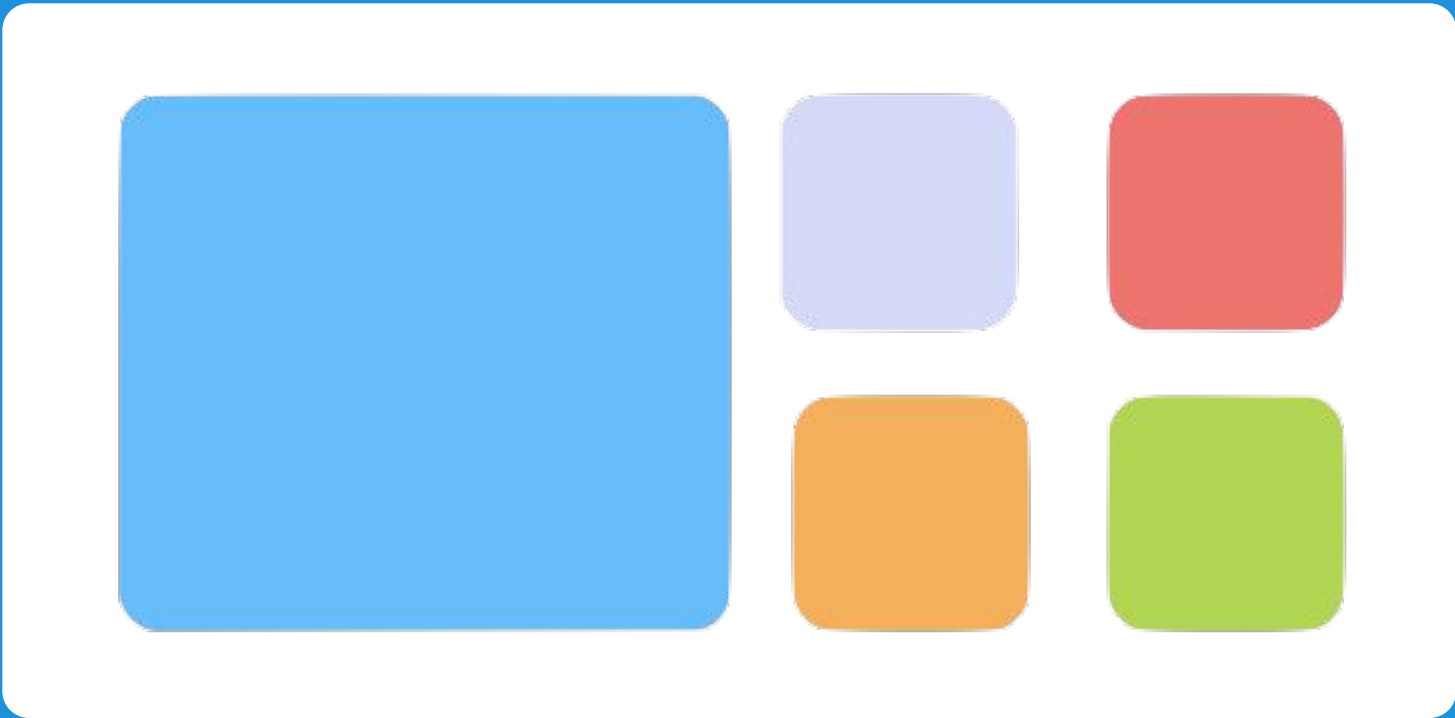
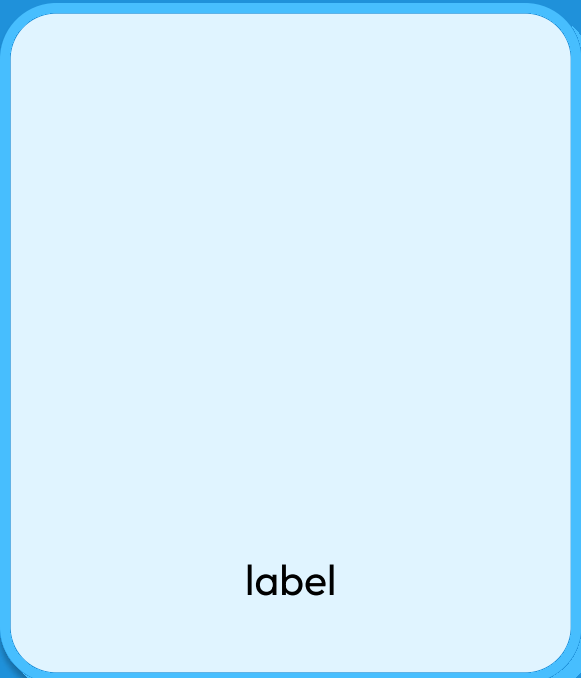
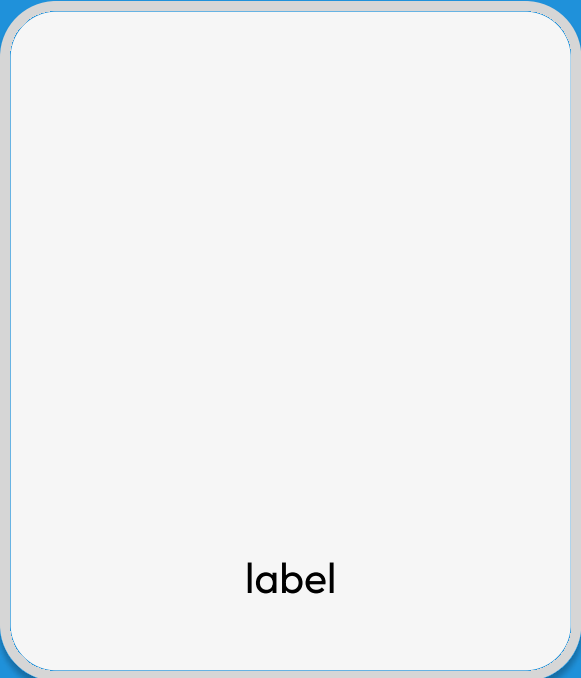
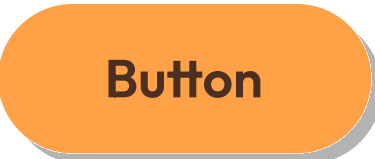
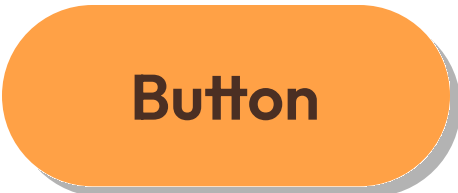
Text Font

Typeface	Weight	Size
H1. Headline	Semi Bold	48
H2. Headline	Semi Bold	40
H3. Headline	Semi Bold	32
H4. Headline	Semi Bold	28
H5. Headline	Semi Bold	24

B1. Body	Regular	24
B1. Body	Medium	24
B2. Body	Regular	14
B2. Body	Medium	14

Button Font

Typeface	Weight	Size
Large	Bold	24
Medium	Bold	16
Small	Bold	12

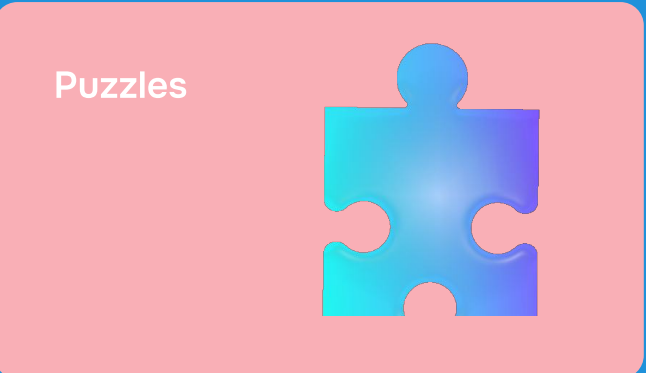
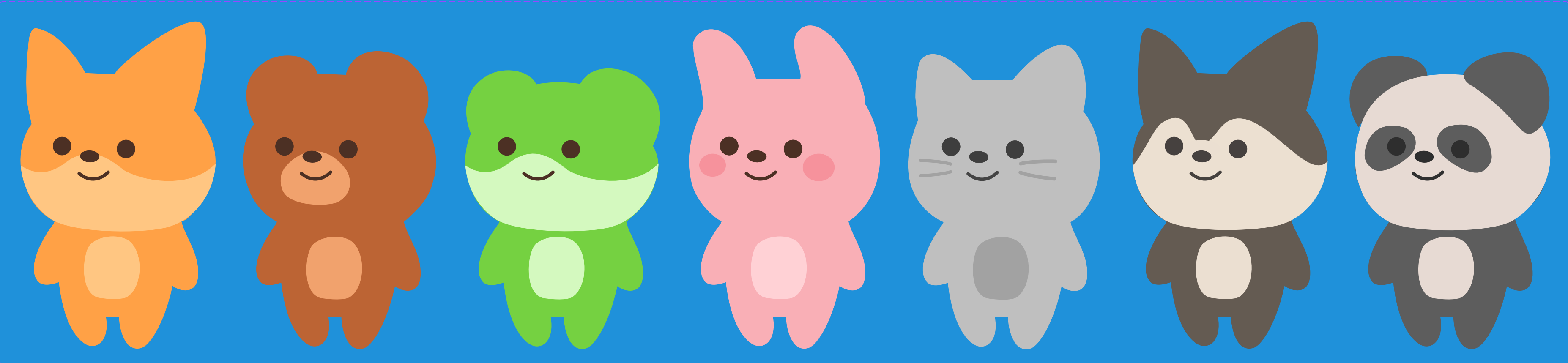


Eddy?



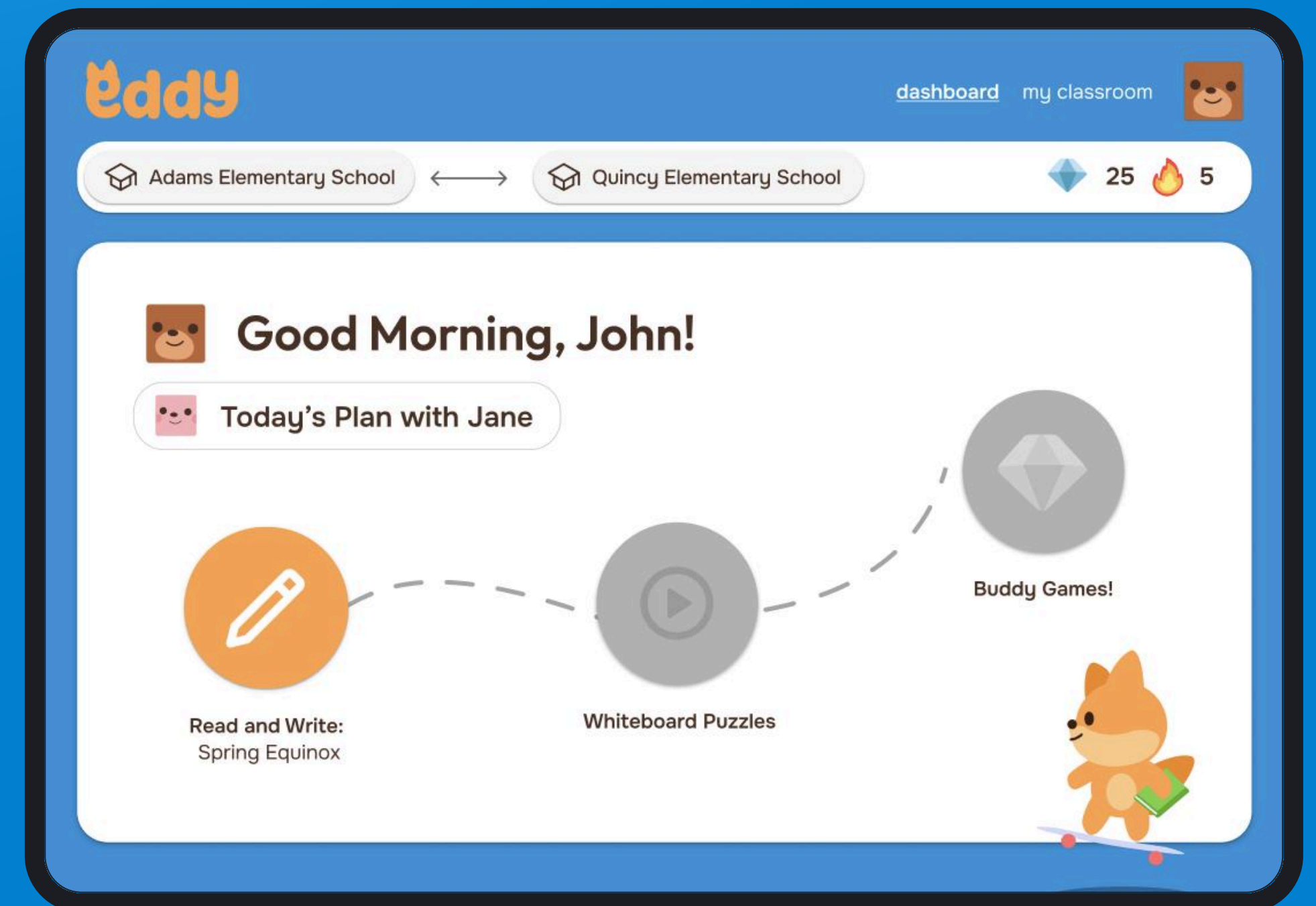
We wanted a fun name to emphasize the youthful and engaging nature of our idea. We chose the name “Eddy” as it combines the beginning of “education”, and personifies our mascot!

02 Ideation



03 Prototype

A demo of our current prototype!



04 Bringing it into the Real World

Project 3: Real world applications of Eddy,
and its anticipated impact



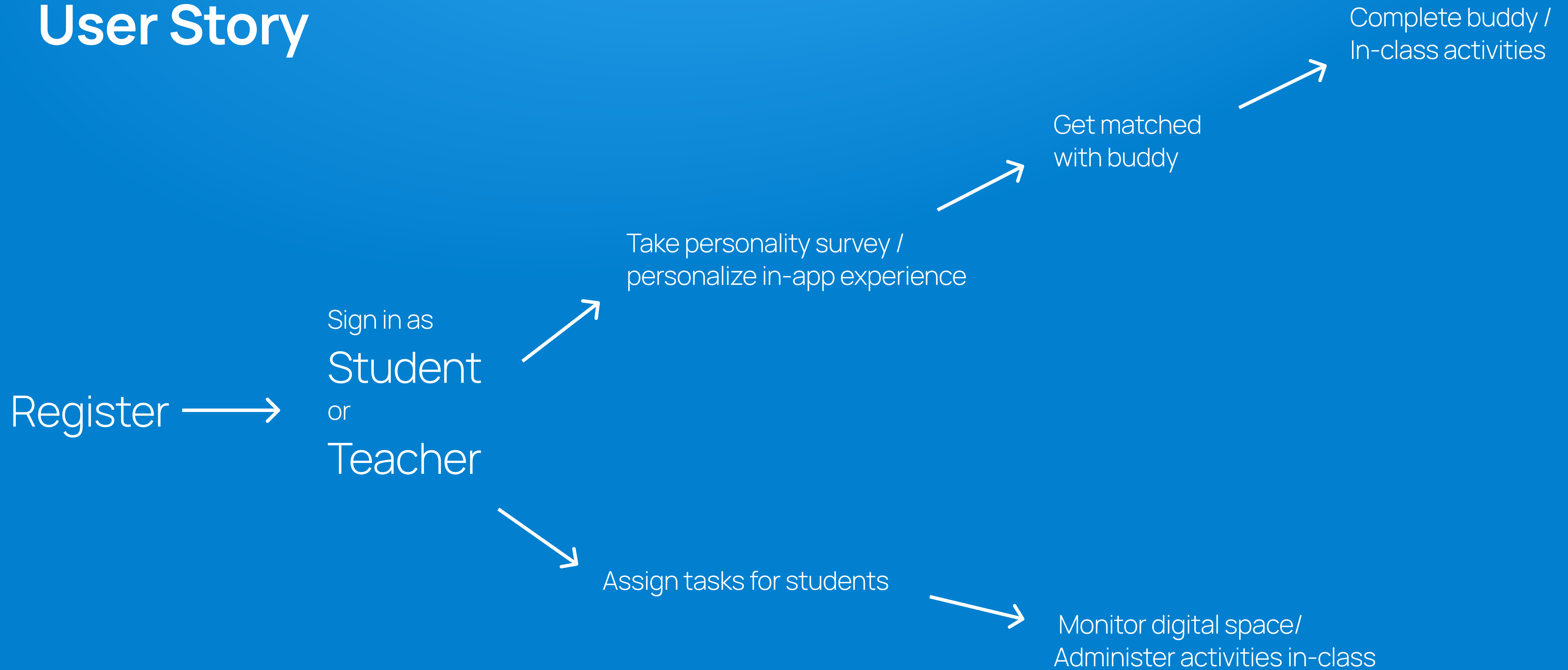


Schools with 1:1 Programs

- Educational program where school provides each student with their own personal learning device (laptop or tablet) that they can take home
- Currently, only 50% of school districts in America have a 1:1 program.



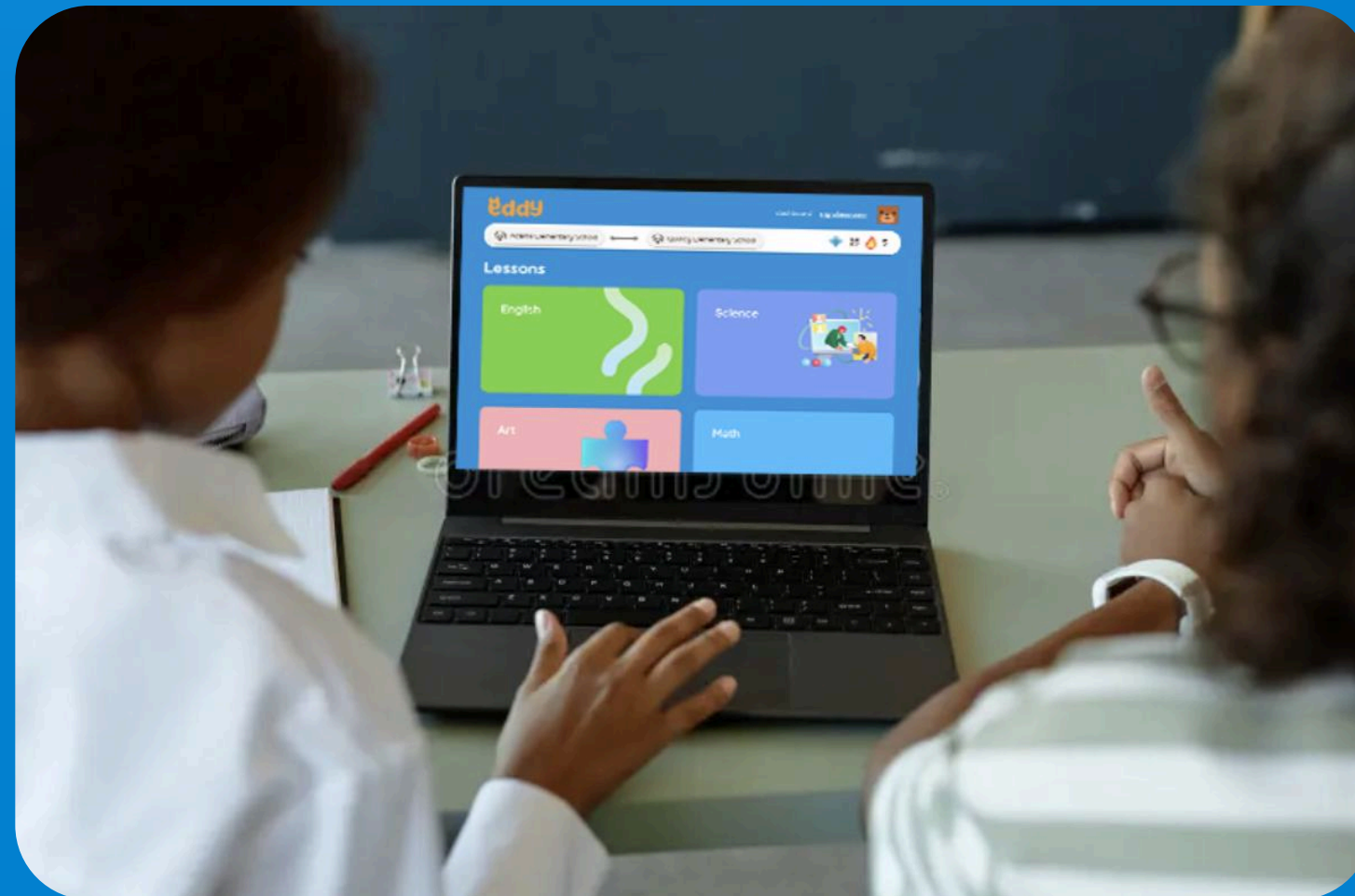
User Story



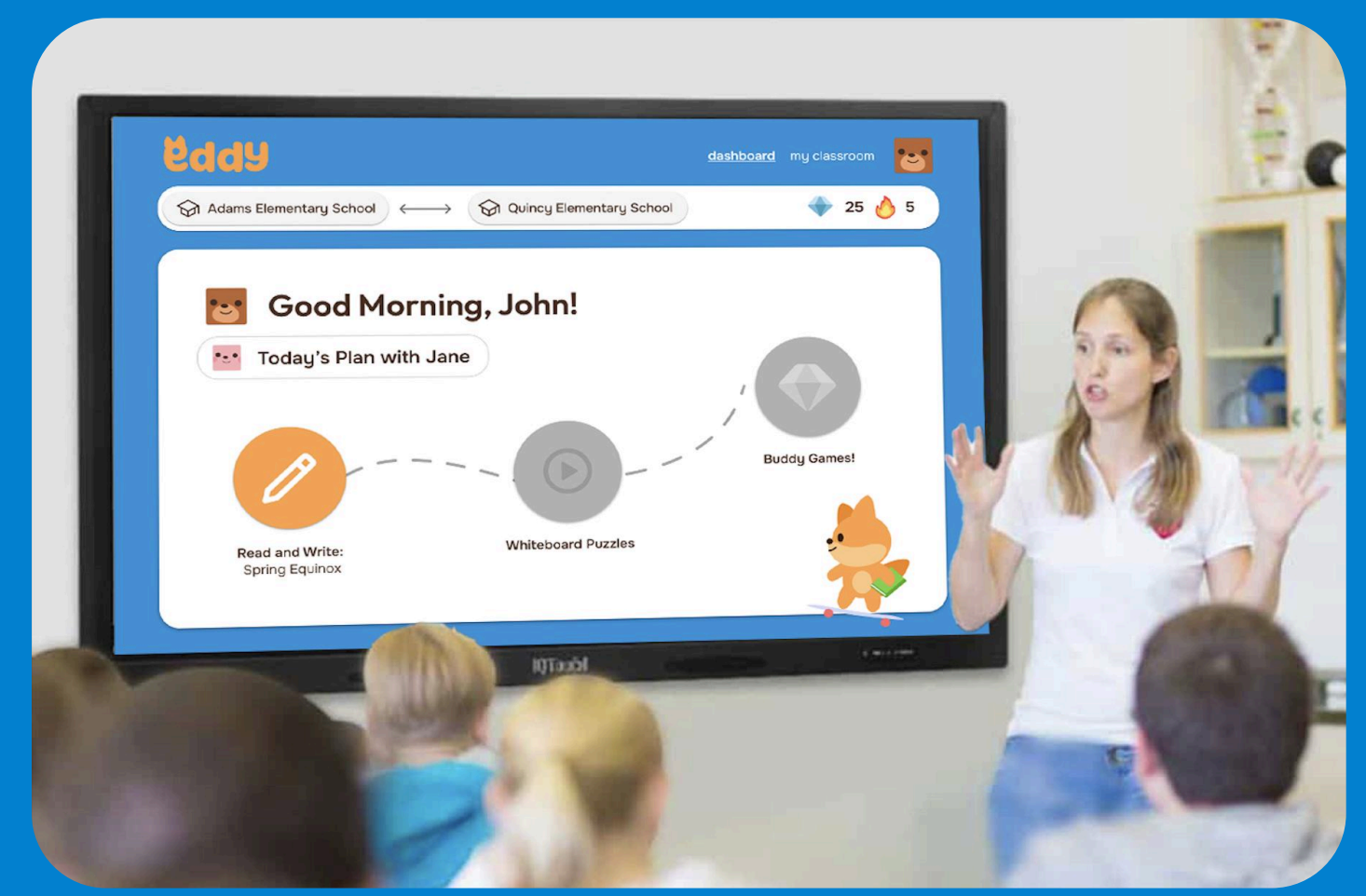
Schools with Less Access to Technology



Computer Labs



Partnering Up



Overhead Projectors/Interactive Whiteboards

Integration into Class

Teachers

1. introduce the new classes to each other and explain the program
2. prepare prompts / learning activities for their students
3. teachers control when students Eddy

Students

1. students first take the quiz and get paired with other class' students
2. students work with each other during a 30-45 minute designated 'Eddy time'



Safety

- Concerns have risen over safety of children on internet, having free access and range online
 - We aim to curb these issues by only allowing access to our program in the classroom and during school hours (controlled by school administrators)
 - Only teacher distribution to students, no outside contact from online users - contained environment only for classroom



Potential Impact

- Students in monocultural or homogenous areas experience children from outside their district in a monitored, equal setting
- Gained social interaction skills with people who are 'different'
- More exposure to academic interests and hobbies that might be discouraged in their community
- Stronger digital communication skills

05 Takeaways

Lessons learned, future goals

Potential Impact

- It is difficult to design for SDGs
 - inclusivity and integration
- learning process
 - many iterations, many ideas



The Future

- Integration with other parts of the country, and the world
- More features for collaboration, student-parent integration
- Students to collaborate in person
- User testing with real elementary school students
- Integration with self-learning modules





Thank You for Listening!

ARTG 2400

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